

Blowup Algebras

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1 What is a Blowup?

We care about studying varieties, and singularities are an obstruction to that. Intuitively, a singularity is a point of the variety where the tangent is not well defined. For instance, in $\mathbb{A}^2$ the curve $y^2 = x^3 + x^2$ has a node at the origin, where two branches cross and there are two perfectly valid tangent directions, $y = x$ and $y = -x$.

Blowups are a systematic tool for resolving singularities like this one. The idea is fairly straightforward: instead of picking one tangent direction at the singular point, we replace the point with a copy of $\mathbb{P}^1$, which parametrizes all lines through it simultaneously. The two branches then lift to the blowup as separate smooth curves, meeting $\mathbb{P}^1$ at different points.

In general, the blowup of a variety $X \subset \mathbb{A}^n$ at a point replaces that point with a copy of $\mathbb{P}^{n-1}$, recording all the directions from which one can approach it. So it lives in $\mathbb{A}^n \times \mathbb{P}^{n-1}$.

Projective space $\mathbb{P}^n$ parametrizes lines through the origin in $\mathbb{A}^{n+1}$: a point $[p_0 : p_1 : \dots : p_n]$ represents the line through 0 and $(p_0, \dots, p_n)$, with two points identified if they differ by a nonzero scalar. A polynomial f therefore defines a well-defined vanishing locus in $\mathbb{P}^n$ only when it is homogeneous, since then $f(\lambda x) = \lambda^d f(x)$ and vanishing is independent of the chosen representative. Subvarieties of $\mathbb{P}^n$ are thus cut out by homogeneous ideals. For the homogeneous coordinate ring $k[x_0, \dots, x_n]$ of $\mathbb{P}^n$, the dictionary sends a homogeneous radical ideal (other than the irrelevant ideal $(x_0, \dots, x_n)$, which gives $\emptyset$) to a projective subvariety; for example,

- (0) — all of $\mathbb{P}^n$ (as a set, $\mathbb{V}(0) = \mathbb{P}^n$);
- (x_i) — a hyperplane $\{x_i = 0\} \cong \mathbb{P}^{n-1}$;
- (x_i, x_j) — a codimension-2 linear subspace $\{x_i = x_j = 0\} \cong \mathbb{P}^{n-2}$;
- (f) for irreducible homogeneous f of degree d — a hypersurface of dimension $n - 1$ (codimension 1, *regardless of d*). This is in general *not* isomorphic to a projective space: a smooth plane cubic ($n = 2, d = 3$) is an elliptic curve, not a $\mathbb{P}^k$.

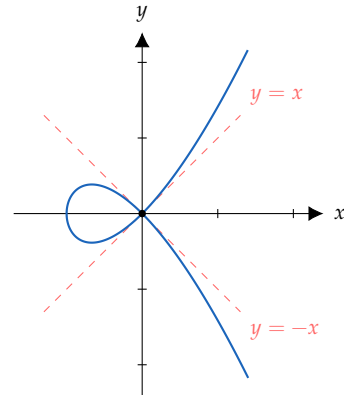


Figure 1: The nodal cubic $y^2 = x^3 + x^2$. The two branches cross at the origin with distinct tangent lines $y = x$ and $y = -x$.

2 Graded Rings and Modules

Since subvarieties of $\mathbb{P}^n$ are cut out by homogeneous ideals, we need a framework to work with them. The polynomial ring $R = k[x_1, \dots, x_n]$ decomposes naturally by degree,

$$R = \bigoplus_{d \geq 0} k[x_1, \dots, x_n]_d,$$

where $R_d = k[x_1, \dots, x_n]_d$ denotes the homogeneous polynomials of degree d . Multiplication respects this decomposition, $R_d \cdot R_e \subseteq R_{d+e}$, and such an $R = \bigoplus_{d \geq 0} R_d$ is called a graded ring. The polynomial ring has a natural grading by degree, but a general ring does not. To extract a graded ring from R , we choose ideals $I_i \subseteq R$ forming a descending multiplicative chain,

$$R \supseteq I_1 \supseteq I_2 \supseteq I_3 \supseteq \dots, \quad I_i I_j \subseteq I_{i+j}.$$

When the I_i are powers of a single ideal, this chain is called an I -adic filtration, and it lets us measure “how deeply” an element lies in I . For our purposes I is always the maximal ideal of a local Noetherian ring R .

From this filtration we recover the associated graded ring $\text{gr}_I R$,

$$\text{gr}_I R := R/I \oplus I/I^2 \oplus I^2/I^3 \oplus \dots,$$

with a well-defined multiplication: for $a \in I^n/I^{n+1}$ and $b \in I^m/I^{m+1}$ we pick representatives $a' \in I^n$ and $b' \in I^m$ and define $ab \in I^{n+m}/I^{n+m+1}$ to be the image of $a'b'$. One checks this is well defined modulo I^{m+n+1} . We now develop some theory around filtrations and graded rings that will be useful later.

The idea of I -adic filtrations generalizes to modules.

Definition 2.1. Given an R -module M and an ideal $I \subseteq R$, we get the I -adic filtration of the module,

$$M \supseteq IM \supseteq I^2 M \supseteq \dots,$$

and the associated graded module,

$$\text{gr}_I M := M/IM \oplus IM/I^2 M \oplus I^2 M/I^3 M \oplus \dots,$$

a graded $\text{gr}_I R$ -module: for $a \in I^n/I^{n+1}$ and $b \in I^m M/I^{m+1} M$ we pick representatives $a' \in I^n$, $b' \in I^m M$ and let $ab \in I^{n+m} M/I^{n+m+1} M$ be the image of $a'b'$.

Remark 2.2. Intersecting the terms of the I -adic filtration of M with a submodule $M' \subseteq M$ generally does not give the I -adic filtration of M' .

The Artin–Rees lemma characterizes exactly when this does hold. We state it now and prove it once the requisite theory is in place.

Lemma 2.3 (Artin–Rees). Let R be a Noetherian ring, $I \subseteq R$ an ideal, and $M' \subseteq M$ finitely generated R -modules. If

$$M = M_0 \supseteq M_1 \supseteq M_2 \supseteq \dots$$

is an I -stable filtration, then the induced filtration

$$M' = M' \cap M_0 \supseteq M' \cap M_1 \supseteq M' \cap M_2 \supseteq \dots$$

is also I -stable; that is, there exists m with $I(M' \cap M_n) = M' \cap M_{n+1}$ for all $n \geq m$.

Definition 2.4. Let R be a ring, $I \subseteq R$ an ideal, and M an R -module. A filtration $M = M_0 \supseteq M_1 \supseteq \cdots$ is an I -filtration if $IM_n \subseteq M_{n+1}$ for all $n \geq 0$, and is I -stable if there exists m with $M_{m+i} = I^i M_m$ for all $i \geq 0$.

Notation. We write M_k for $I^k M$, and

$$\mathrm{gr}_I R = \bigoplus_{k \geq 0} I^k / I^{k+1}, \quad \mathrm{gr}_I M = \bigoplus_{k \geq 0} M_k / M_{k+1}$$

for the associated graded ring and module. We also need the *Rees algebra* (studied in detail in §3) and *Rees module*: for an I -filtration $\mathcal{J} : M = M_0 \supseteq M_1 \supseteq \cdots$,

$$B_I R := \bigoplus_{n \geq 0} I^n t^n \subseteq R[t], \quad B_{\mathcal{J}} M := \bigoplus_{n \geq 0} M_n t^n,$$

where $B_{\mathcal{J}} M$ is a graded $B_I R$ -module. (When $\mathcal{J}$ is the I -adic filtration, $M_n = I^n M$.)

There is no canonical ring homomorphism $R \rightarrow \mathrm{gr}_I R$ lifting the structure of R . The only canonical ring map out of R compatible with the grading is the projection onto degree 0,

$$R \twoheadrightarrow R/I = (\mathrm{gr}_I R)_0 \hookrightarrow \mathrm{gr}_I R,$$

which forgets everything in positive degree. The genuinely useful gadget attached to the associated graded is instead a *set* map (not a ring homomorphism), the initial form.

Definition 2.5. Let $f \in M$ and let m be the greatest integer with $f \in M_m$. The initial form of f is

$$\mathrm{in}(f) = f \pmod{M_{m+1}} \in M_m / M_{m+1} \subseteq \mathrm{gr}_I M,$$

or $\mathrm{in}(f) = 0$ if $f \in \bigcap_m M_m$.

Definition 2.6. For a submodule $M' \subseteq M$, define $\mathrm{in}(M')$ to be the $\mathrm{gr}_I R$ -submodule of $\mathrm{gr}_I M$ generated by $\mathrm{in}(f)$ for all $f \in M'$.

Remark 2.7. The initial-form map is not additive, and it is not multiplicative when $\mathrm{gr}_I R$ has zero-divisors; in particular $\mathrm{in}(M')$ is generally not the submodule generated by the initial forms of a generating set of M' .

For instance, take $R = k[x, y]$, $I = (x, y)$, and $J = (x^2 + y^3, xy) \subseteq R$ with the I -adic filtration. The initial forms of the generators are $\mathrm{in}(x^2 + y^3) = x^2$ and $\mathrm{in}(xy) = xy$, so they generate $(x^2, xy) \subseteq \mathrm{gr}_I R \cong k[x, y]$. However,

$$y \cdot (x^2 + y^3) - x \cdot (xy) = x^2 y + y^4 - x^2 y = y^4 \in J,$$

so $\mathrm{in}(y^4) = y^4 \in \mathrm{in}(J)$. But $y^4 \notin (x^2, xy)$, since every element of (x^2, xy) has x as a factor while y^4 does not. Hence $\mathrm{in}(J) \supsetneq (\mathrm{in}(x^2 + y^3), \mathrm{in}(xy))$.

Proposition 2.8. Let R be a ring, $I \subseteq R$ an ideal, and M a finitely generated R -module with an I -filtration $\mathcal{J} : M = M_0 \supseteq M_1 \supseteq \cdots$ by finitely generated modules M_i . Then $\mathcal{J}$ is I -stable if and only if the $B_I R$ -module $B_{\mathcal{J}} M$ is finitely generated.

Proof. ($\Rightarrow$) Assume $\mathcal{J}$ is I -stable, so $M_{n+i} = I^i M_n$ for some n and all $i \geq 0$; that is, for $k > n$, M_k is obtained from M_n by multiplying by powers of I . Since each of $M_0, \dots, M_n$ is finitely generated,

the union of their generating sets (placed in their respective degrees) gives a finite generating set for $B_{\mathcal{J}}M$.

($\Leftarrow$) Assume $B_{\mathcal{J}}M$ is finitely generated. Since $B_{\mathcal{J}}M = \bigoplus_{n \geq 0} M_n t^n$ is a direct sum, each generator has only finitely many nonzero homogeneous components, so each generator lies in $M_0 \oplus \cdots \oplus M_{k_i}$ for some k_i . Replacing the generators by their homogeneous pieces (which still generate), we may assume $B_{\mathcal{J}}M$ is generated by homogeneous elements lying in degrees $\leq n$, where $n = \max_i k_i$.

Now take any element of M_{n+i} for $i \geq 0$. As a degree- $(n+i)$ element of $B_{\mathcal{J}}M$, it is a sum of terms $a \cdot m$ with m a generator in some degree $k \leq n$ and $a \in (B_I R)_{n+i-k} = I^{n+i-k}$. Write $a \cdot m = I^i \cdot (I^{n-k}m)$; since $\mathcal{J}$ is an I -filtration, $I^{n-k}m \in M_n$, so $a \cdot m \in I^i M_n$. Hence $M_{n+i} \subseteq I^i M_n$. The reverse inclusion is immediate, so $M_{n+i} = I^i M_n$ for all $i \geq 0$ and $\mathcal{J}$ is I -stable. $\square$

Proof of Artin–Rees. Let

$$\mathcal{J}' : M' = M'_0 \supseteq M'_1 := M' \cap M_1 \supseteq M'_2 := M' \cap M_2 \supseteq \cdots$$

be the induced filtration on M' . Then $B_{\mathcal{J}'}M' = \bigoplus_{n \geq 0} (M' \cap M_n) t^n$ is a graded $B_I R$ -submodule of $B_{\mathcal{J}}M$.

Since $\mathcal{J}$ is I -stable, $B_{\mathcal{J}}M$ is finitely generated as a $B_I R$ -module by Proposition 2.8. As R is Noetherian and I is finitely generated, $B_I R = \bigoplus_{n \geq 0} I^n t^n$ is a finitely generated R -algebra, hence a quotient of a polynomial ring over R and so Noetherian by the Hilbert basis theorem. Therefore the submodule $B_{\mathcal{J}'}M'$ of the finitely generated $B_I R$ -module $B_{\mathcal{J}}M$ is itself finitely generated, and Proposition 2.8 (applied to M') shows $\mathcal{J}'$ is I -stable. $\square$

Remark 2.9. A key payoff of the associated graded construction: when $I = \mathfrak{m}$ is the maximal ideal of a Noetherian local ring, $\text{gr}_{\mathfrak{m}} R$ is a finitely generated graded algebra over the field $R/\mathfrak{m}$, i.e. a quotient of a polynomial ring by a homogeneous ideal. This matters because such algebras are exactly the homogeneous coordinate rings of projective schemes, so passing to $\text{gr}_{\mathfrak{m}} R$ lets us deploy the full toolkit of projective geometry — Proj, Hilbert functions, Hilbert polynomials, dimension, multiplicity — on an object built from an otherwise unstructured local ring. Crucially, the passage $R \rightsquigarrow \text{gr}_{\mathfrak{m}} R$ preserves the local numerical invariants we care about: $\dim R = \dim \text{gr}_{\mathfrak{m}} R$, and the Hilbert–Samuel function $n \mapsto \ell(R/\mathfrak{m}^n)$ of R is computed by the Hilbert function of $\text{gr}_{\mathfrak{m}} R$. So dimension and multiplicity can be read off the much more tractable graded ring, and geometrically $\text{Spec } \text{gr}_{\mathfrak{m}} R$ is the tangent cone — the linearization of $\text{Spec } R$ at the point.

3 Blowup Algebras

Definition 3.1. Let R be a ring and $I \subseteq R$ an ideal. The **blowup algebra** (Rees algebra) of I in R is the R -algebra

$$B_I R := \bigoplus_{n \geq 0} I^n t^n = R \oplus It \oplus I^2 t^2 \oplus \cdots \cong R[It] \subseteq R[t],$$

where t is a formal variable tracking the power of I . The associated graded ring is obtained by quotienting by $IB_I R$:

$$B_I R / IB_I R = \text{gr}_I R = R/I \oplus I/I^2 \oplus \cdots.$$

The blowup algebra is the algebraic structure behind the geometric operation of blowing up a subvariety. Suppose R is the coordinate k -algebra of an affine variety X , and $I \subseteq R$ is the ideal of a subvariety $Y \subseteq X$. Let $a_1, \dots, a_r$ be k -algebra generators for R and $g_0, \dots, g_s$ generators of I as an ideal of R .

Proposition 3.2. The blowup algebra $B_I R$ is a homomorphic image of $k[x_1, \dots, x_r, y_0, \dots, y_s]$.

Proof. Define a ring homomorphism

$$\phi : k[x_1, \dots, x_r, y_0, \dots, y_s] \rightarrow B_I R, \quad x_i \mapsto a_i, \quad y_j \mapsto g_j t.$$

This is surjective: any element of $B_I R$ is a finite sum of terms $f \cdot g t^n$ with $f \in R$, $g \in I^n$, and since f is a polynomial in the a_i and g is a polynomial in the g_j , every such term is in the image. By the first isomorphism theorem,

$$B_I R \cong k[x_1, \dots, x_r, y_0, \dots, y_s] / \ker(\phi).$$

Since $B_I R$ is graded (by the power of t) and ϕ respects the grading, sending the x_i to degree 0 and the y_j to degree 1, the kernel is a homogeneous ideal. $\square$

Here $k[x_1, \dots, x_r]$ is the coordinate ring of $\mathbb{A}^r$ and $k[y_0, \dots, y_s]$ is the homogeneous coordinate ring of $\mathbb{P}^s$, so $k[x_1, \dots, x_r, y_0, \dots, y_s]$ is the coordinate ring of $\mathbb{A}^r \times \mathbb{A}^{s+1}$. Quotienting by a homogeneous ideal in the y_j makes vanishing scale-invariant in the y_j and cuts out a subvariety that is projective in the y -directions,

$$Z = \mathbb{V}(\ker \phi) \subseteq \mathbb{A}^r \times \mathbb{P}^s.$$

The projection $\pi : \mathbb{A}^r \times \mathbb{P}^s \rightarrow \mathbb{A}^r$ restricts to a map $Z \rightarrow X$ which is an isomorphism away from the preimage of Y . The set $Z = \text{Proj}(B_I R)$ is the **blowup of Y in X** , and $\pi^{-1}(Y) \subseteq Z$ is the **exceptional set**.

Algebraically, the exceptional set comes from killing I inside the blowup algebra: $B_I R / IB_I R = \text{gr}_I R$, so

$$\pi^{-1}(Y) = \text{Proj}(\text{gr}_I R) \subseteq \mathbb{P}^s.$$

Recall that $\text{gr}_I R = \bigoplus_{n \geq 0} I^n / I^{n+1}$ is a graded ring, and its homogeneous primes not containing the irrelevant ideal $(\text{gr}_I R)_{>0}$ correspond to points of a projective variety, exactly as for $k[y_0, \dots, y_s]$ and $\mathbb{P}^s$. This variety — the projectivized normal cone of Y in X — parametrizes the limiting directions of approach to Y inside X : when Y is the origin in $X \subseteq \mathbb{A}^r$ it is the projectivized tangent cone, with one point for each limiting direction of a secant line to X through Y . We make this precise in §4.

We can now revisit the example from the start of the note more formally.

Example: blowing up the node of $y^2 = x^3 + x^2$

The curve $X = \mathbb{V}(y^2 - x^3 - x^2)$ has a node at the origin. With $f = y^2 - x^3 - x^2$, the Jacobian $(\partial_x f, \partial_y f) = (-3x^2 - 2x, 2y)$ vanishes at $(0, 0)$, and $f(0, 0) = 0$, so the origin is singular.

Blow up $Y = \{0\}$ inside X . The coordinate ring is $R = k[x, y]/(y^2 - x^3 - x^2)$, and the maximal ideal of the origin is $I = (x, y)$. Both R (as a k -algebra) and I (as an ideal) are generated by x, y , so by Proposition 3.2 the blowup algebra is the image of $k[x_1, x_2, y_0, y_1]$ under

$$\phi : x_1 \mapsto x, \quad x_2 \mapsto y, \quad y_0 \mapsto xt, \quad y_1 \mapsto yt.$$

Two relations are visibly in $\ker \phi$: the *curve relation* $F = x_2^2 - x_1^3 - x_1^2$, and the *incidence relation* $G = x_1 y_1 - x_2 y_0$, since $\phi(G) = x \cdot yt - y \cdot xt = 0$. Both are homogeneous of degree ≤ 1 in y_0, y_1 , so they cut out a subset of $\mathbb{A}^2 \times \mathbb{P}^1$.

Remark 3.3 (A subtlety worth flagging). F and G do not generate $\ker \phi$, and $\mathbb{V}(F, G)$ is not the blowup. Setting $x = y = 0$ kills both F and G , so $\mathbb{V}(F, G)$ contains the entire exceptional line $\{0\} \times \mathbb{P}^1$; in fact $\mathbb{V}(F, G) = \pi^{-1}(X)$ is the reducible total transform $\tilde{X} \cup (\{0\} \times \mathbb{P}^1)$. The blowup $\text{Bl}_0 X = \text{Proj}(B_I R) = \tilde{X}$ is the irreducible strict transform: the unique component dominating X , obtained by saturating (F, G) with respect to the irrelevant ideal (y_0, y_1) (i.e. removing the exceptional component). Concretely, $\ker \phi$ contains further relations; for example

$$\phi(y_1^2 - (1 + x_1)y_0^2) = (yt)^2 - (1 + x)(xt)^2 = (y^2 - x^2 - x^3)t^2 = 0,$$

and $y_1^2 - (1 + x_1)y_0^2$ does not vanish on all of $\{0\} \times \mathbb{P}^1$ — it cuts the exceptional line down to the points that genuinely lie on the blowup.

It is cleanest to compute the strict transform in the two standard charts of $\text{Bl}_0 \mathbb{A}^2 = \mathbb{V}(xv - yu) \subseteq \mathbb{A}^2 \times \mathbb{P}^1_{[u:v]}$ (here $[u : v]$ are the coordinates with $u \leftrightarrow xt, v \leftrightarrow yt$, so $[u : v] = [x : y]$ off the origin):

- *Chart $u \neq 0$.* Set $u = 1$ and let $m = v$, so $y = mx$. Substituting into $y^2 = x^3 + x^2$ gives

$$m^2 x^2 = x^2(x + 1) \implies x^2(m^2 - x - 1) = 0.$$

The factor x^2 is the exceptional divisor; the strict transform is $m^2 = x + 1$, i.e. $x = m^2 - 1$, $y = m^3 - m$ — a smooth curve parametrized by m . It meets the exceptional divisor $x = 0$ at $m = \pm 1$.

- *Chart $v \neq 0$.* Symmetrically, with $n = u/v$ and $x = ny$ one gets $y^2(1 - n^2 - n^3 y) = 0$; the strict transform meets the exceptional divisor at the same two points (and avoids the vertical direction $[0 : 1]$).

So blowing up separates the branches: the strict transform is smooth and meets the exceptional $\mathbb{P}^1$ in the two points $[u : v] = [1 : 1]$ and $[1 : -1]$, i.e. the directions $y = x$ and $y = -x$. These two points are exactly the exceptional set $\pi^{-1}(0) \cap \tilde{X}$, which we identify with $\text{Proj}(\text{gr}_I R)$ in §4.

Algebraically, the exceptional set is what remains after setting the affine coordinates to zero: it is the projective variety in $\mathbb{P}^s$ cut out by the relations of $\ker \phi$ with $x_1 = \cdots = x_r = 0$. Equivalently, killing I in the blowup algebra leaves $B_I R / IB_I R = \text{gr}_I R$, and the exceptional set is the projective variety defined by these (homogeneous) relations. This variety (the projectivized normal cone of Y in X) parametrizes the limiting directions of approach to Y inside X : when Y is the origin in $X \subseteq \mathbb{A}^r$ it is the projectivized tangent cone, with one point for each limiting direction of a secant line to X through Y . We make this precise in Section 4.

Charts of the blowup

We realized the blowup concretely as the subvariety $Z = \mathbb{V}(\ker \phi) \subseteq \mathbb{A}^r \times \mathbb{P}^s$. To compute with it we cover it by affine pieces, exactly as one covers $\mathbb{P}^s$ by the standard charts $\{y_j \neq 0\}$. We spell this out for a point in affine space, the case we need.

Take $X = \mathbb{A}^n$ with coordinates $x_1, \dots, x_n$ and $Y = \{0\}$, so $I = (x_1, \dots, x_n)$. The blowup sits in $\mathbb{A}^n \times \mathbb{P}^{n-1}$, with $\mathbb{P}^{n-1}$ carrying coordinates $[u_1 : \cdots : u_n]$ (here $u_i \leftrightarrow x_i t$). The relations $x_i u_j = x_j u_i$ say that the point $(x_1, \dots, x_n)$ and the direction $(u_1, \dots, u_n)$ are proportional, so the blowup of $\mathbb{A}^n$ at the origin, denoted as $\text{Bl}_0 \mathbb{A}^n$, is the incidence variety of points and the lines through them,

$$\text{Bl}_0 \mathbb{A}^n = \{(p, [\ell]) \in \mathbb{A}^n \times \mathbb{P}^{n-1} : p \in \ell\} = \mathbb{V}(x_i u_j - x_j u_i : i < j) \subseteq \mathbb{A}^n \times \mathbb{P}^{n-1}.$$

(For $X = \mathbb{A}^n$ these incidence relations already generate $\ker \phi$: since $\mathbb{A}^n$ is smooth there is no exceptional component to discard, in contrast to the node below.)

Definition 3.4 (Affine charts). *On the standard chart $U_i = \{u_i \neq 0\}$ of $\mathbb{P}^{n-1}$, dehomogenize by setting $u_i = 1$ and writing $m_j = u_j / u_i$ for $j \neq i$. The relations $x_i u_j = x_j u_i$ become $x_j = m_j x_i$, so every coordinate is a polynomial in x_i and the m_j . Hence the i -th chart of $\text{Bl}_0 \mathbb{A}^n$ is itself an affine space,*

$$\text{Bl}_0 \mathbb{A}^n \cap (\mathbb{A}^n \times U_i) \cong \mathbb{A}^n \text{ with coordinates } (x_i, m_j)_{j \neq i},$$

and in these coordinates the projection π is

$$\pi(x_i, (m_j)_{j \neq i}) = (x_1, \dots, x_n), \quad x_j = m_j x_i \ (j \neq i).$$

In this chart the exceptional divisor $E = \pi^{-1}(0)$ is the coordinate hyperplane $\{x_i = 0\}$, and the n charts glue along E to recover $E \cong \mathbb{P}^{n-1}$.

For $n = 2$, writing $[u : v]$ for $\mathbb{P}^1$, there are two charts:

$$\{u \neq 0\} : m = \frac{v}{u}, \ y = mx, \ \text{coords } (x, m), \ \pi(x, m) = (x, mx), \ E = \{x = 0\};$$

$$\{v \neq 0\} : n = \frac{u}{v}, \ x = ny, \ \text{coords } (y, n), \ \pi(y, n) = (ny, y), \ E = \{y = 0\}.$$

To blow up a plane curve $X = \mathbb{V}(f) \subseteq \mathbb{A}^2$ through the origin, we pull f back to each chart. Write $f = f_{m_0} + f_{m_0+1} + \cdots$ in homogeneous parts with $f_{m_0} = \text{in}(f) \neq 0$, so $m_0 = \text{ord}_0(f)$ is the multiplicity of X at 0. In the chart $\{u \neq 0\}$, since $f_d(x, mx) = x^d f_d(1, m)$,

$$f(x, mx) = \sum_{d \geq m_0} x^d f_d(1, m) = x^{m_0} \left(\underbrace{\text{in}(f)(1, m)}_{= f_{m_0}(1, m)} + x(\cdots) \right).$$

The factor x^{m_0} is the exceptional divisor taken with multiplicity m_0 — the part of the pullback supported over 0, giving the *total transform* $\pi^{-1}(X)$ — and the remaining factor $\tilde{f}(x, m)$ cuts out the *strict transform*, the closure $\tilde{X} = \overline{\pi^{-1}(X \setminus 0)}$. Setting $x = 0$, the strict transform meets the exceptional divisor exactly where

$$\text{in}(f)(1, m) = 0,$$

i.e. at the projectivized zeros of the initial form: the tangent directions of X at 0. This is the geometric shadow of the tangent cone theorem of §4.

Blowing up the node of $y^2 = x^3 + x^2$

The curve $X = \mathbb{V}(f)$ with $f = y^2 - x^3 - x^2$ has a node at the origin: the Jacobian $(\partial_x f, \partial_y f) = (-3x^2 - 2x, 2y)$ vanishes at $(0, 0)$ and $f(0, 0) = 0$, and the lowest-degree part $\text{in}(f) = y^2 - x^2 = (y - x)(y + x)$ has two distinct linear factors — multiplicity $m_0 = 2$, with tangent directions $y = \pm x$.

We blow up $Y = \{0\}$. The coordinate ring is $R = k[x, y] / \langle y^2 - x^3 - x^2 \rangle$ and the ideal of the origin is $I = \langle x, y \rangle$ (by the Nullstellensatz). Both R and I are generated by x, y , so by Proposition ?? the blowup algebra is the image of $k[x_1, x_2, y_0, y_1]$ under

$$\phi : \begin{cases} x_1 \mapsto x, \\ x_2 \mapsto y, \\ y_0 \mapsto xt, \\ y_1 \mapsto yt. \end{cases}$$

Two relations sit visibly in $\ker \phi$: the curve relation $F = x_2^2 - x_1^3 - x_1^2$ and the incidence relation $G = x_1 y_1 - x_2 y_0$ (since $\phi(G) = x \cdot yt - y \cdot xt = 0$). These cut out a set $Z \subseteq \mathbb{A}^2 \times \mathbb{P}^1$ — but F and G both vanish at $x = y = 0$, so $\mathbb{V}(F, G)$ contains the entire exceptional line $\{0\} \times \mathbb{P}^1$ and is the reducible total transform.

Eliminating t as in Remark ?? gives the full kernel,

$$\ker \phi = (F, G, H, P), \quad H = x_2 y_1 - (x_1^2 + x_1) y_0, \quad P = y_1^2 - (1 + x_1) y_0^2,$$

(both H and P check out: $\phi(H) = (y^2 - x^3 - x^2)t = 0$ and $\phi(P) = (y^2 - x^2 - x^3)t^2 = 0$). The two extra generators are what trim the exceptional line off the total transform: restricted to $\{0\} \times \mathbb{P}^1$, the relation P becomes $y_1^2 - y_0^2$, which vanishes only at $[y_0 : y_1] = [1 : 1]$ and $[1 : -1]$. The same two points fall out — more transparently — from the charts of Definition 3.4.

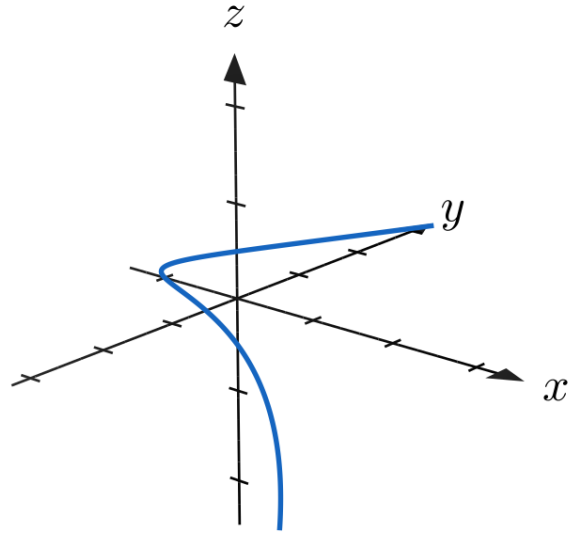


Figure 2: The strict transform of the node in the slope chart $\mathbb{A}^2_{x,y} \times \mathbb{A}^1_z$ of the blowup, where the vertical coordinate is the slope $z = y/x$. The smooth space curve $z \mapsto (z^2 - 1, z^3 - z, z)$ has no self-intersection; projecting it to the xy -plane (forgetting z) collapses it onto the nodal cubic, regluing the points $z = +1$ and $z = -1$ at the origin. Blowing up has lifted the two branches to the heights $z = \pm 1$, the tangent slopes $y = \pm x$.

Chart $\{u \neq 0\}$ (slope $m = v/u$, so $y = mx$). Substituting,

$$f(x, mx) = (mx)^2 - x^3 - x^2 = x^2(m^2 - x - 1).$$

The factor x^2 is the exceptional divisor (multiplicity $m_0 = 2$); discarding it, the strict transform is the closure of the locus with $x \neq 0$, namely

$$\tilde{X}: m^2 - x - 1 = 0, \quad \text{i.e.} \quad x = m^2 - 1, \quad y = mx = m^3 - m,$$

a smooth curve. It meets the exceptional divisor $x = 0$ where $m^2 - 1 = 0$, i.e. at $m = \pm 1$ — the directions $[u : v] = [1 : 1]$ and $[1 : -1]$, exactly the roots of $\text{in}(f)(1, m) = m^2 - 1$.

Chart $\{v \neq 0\}$ ($n = u/v$, so $x = ny$). Substituting,

$$f(ny, y) = y^2 - n^3 y^3 - n^2 y^2 = y^2(1 - n^2 - n^3 y),$$

with strict transform $1 - n^2 - n^3 y = 0$. It meets $y = 0$ where $1 - n^2 = 0$, i.e. at $n = \pm 1$ — the same two points $[1 : \pm 1]$ — and misses the vertical direction $[0 : 1]$ (there $1 = 0$ has no solution).

So blowing up separates the branches: the strict transform $\tilde{X}$ is smooth, and it meets the exceptional $\mathbb{P}^1$ in the two points $[1 : 1]$ and $[1 : -1]$, i.e. the tangent directions $y = \pm x$. These two points are the exceptional set $\pi^{-1}(0) \cap \tilde{X}$, which we identify with the projectivized tangent cone in §4.

4 Tangent Cone

Theorem 4.1 (Krull's Intersection Theorem). *Let R be a Noetherian ring, $I \subseteq R$ an ideal, and M a finitely generated R -module. There exists $r \in I$ with*

$$(1 - r) \bigcap_{k=1}^{\infty} I^k M = 0.$$

Proof. Let $N = \bigcap_{k=1}^{\infty} I^k M$; we find $r \in I$ with $(1 - r)N = 0$. By Artin–Rees applied to $N \subseteq M$ (with the I -adic filtration on M), there exists m such that for all $n \geq m$,

$$N \cap I^n M = I^{n-m}(N \cap I^m M).$$

In particular $N \cap I^{m+1} M = I(N \cap I^m M)$. But $N \subseteq I^{m+1} M$ and $N \subseteq I^m M$, so $N \cap I^{m+1} M = N$ and $N \cap I^m M = N$, giving $N = IN$. Since R is Noetherian and M is finitely generated, N is finitely generated, and Nakayama's lemma applied to $N = IN$ yields $r \in I$ with $(1 - r)N = 0$. $\square$

Corollary 4.1.1. *Let R be Noetherian. If R is a domain, or $(R, \mathfrak{m})$ is local, and $I \subsetneq R$, then $\bigcap_{k=1}^{\infty} I^k = 0$.*

Proof. Take $M = R$, so there is $r \in I$ with $(1 - r) \bigcap_{k=1}^{\infty} I^k = 0$. As I is proper, $1 \notin I$, so $r \neq 1$ and $1 - r \neq 0$. If R is a domain this forces $\bigcap_{k=1}^{\infty} I^k = 0$. If R is local, $r \in I \subseteq \mathfrak{m}$, so $1 - r \notin \mathfrak{m}$ is a unit, again forcing $\bigcap_{k=1}^{\infty} I^k = 0$. $\square$

Corollary 4.1.2. *Let R be a Noetherian local ring and $I \subsetneq R$ an ideal. If $\text{gr}_I R$ is a domain, then so is R .*

Proof. Suppose $fg = 0$ in R with $f, g \neq 0$. The product $\text{in}(f)\text{in}(g)$ is, by definition, the image of fg in the relevant graded piece, hence $\text{in}(f)\text{in}(g) = 0$ in $\text{gr}_I R$. As $\text{gr}_I R$ is a domain, $\text{in}(f) = 0$ or $\text{in}(g) = 0$; say $\text{in}(f) = 0$. Then $f \in \bigcap_k I^k = 0$ by the previous corollary, so $f = 0$, a contradiction. Hence R is a domain. $\square$

We now make the link between the exceptional set and the associated graded ring geometric. Throughout, $R = k[x_1, \dots, x_r]/J$ with $I = (x_1, \dots, x_r)$, k algebraically closed, and $J \subseteq I$ (so the origin lies on X); let $X = \mathbb{V}(J) \subseteq \mathbb{A}^r$.

Definition 4.2. A point $p = (p_1, \dots, p_r) \in \mathbb{A}^r \setminus \{0\}$ spans a secant line $\overline{0p}$, i.e. a direction $[p_1 : \dots : p_r] \in \mathbb{P}^{r-1}$. The set of limiting secant directions,

$$\text{PTC}_0(X) := \overline{\{[p] : p \in X \setminus \{0\}\}} \subseteq \mathbb{P}^{r-1},$$

is the **projectivized tangent cone**. Over $\mathbb{C}$ the closure is genuinely a set of limits: if $p_n \in X \setminus \{0\}$ with $p_n \rightarrow 0$, the directions $[p_n]$ have a convergent subsequence because $\mathbb{P}^{r-1}$ is compact. The **(affine) tangent cone** $\text{TC}_0(X) \subseteq \mathbb{A}^r$ is the cone over $\text{PTC}_0(X)$ — the union of all these lines through the origin (together with 0). It is the affine version, and the object the theorem below describes; its projectivization $\text{PTC}_0(X)$ is precisely the exceptional set of $\text{Bl}_0 X$ from §3.

One shows that the affine tangent cone is cut out by the initial ideal $\text{in}_I(J)$ and that its coordinate ring is $\text{gr}_I R$; this is the content of the next theorem. (For the general theory of tangent cones, associated graded rings, and the Artin–Rees lemma, see Eisenbud, *Commutative Algebra with a View Toward Algebraic Geometry*, Ch. 5, and Harris, *Algebraic Geometry: A First Course*.)

Theorem 4.3. Let $R = k[x_1, \dots, x_r]/J$ with $I = (x_1, \dots, x_r)$, k algebraically closed, and $J \subseteq I$. The (affine) tangent cone to $X = \mathbb{V}(J) \subseteq \mathbb{A}^r$ at the origin is the algebraic set

$$\text{TC}_0(X) = \mathbb{V}(\text{in}_I(J)) \subseteq \mathbb{A}^r,$$

where $\text{in}_I(J) = \langle \text{in}(f) : f \in J \rangle \subseteq k[x_1, \dots, x_r]$ is the ideal generated by the initial forms of all elements of J . Moreover, the coordinate ring of the tangent cone is

$$k[x_1, \dots, x_r]/\text{in}_I(J) \cong \text{gr}_I R.$$

Proof. Let $S = k[x_1, \dots, x_r]$ and $\mathfrak{m} = (x_1, \dots, x_r) \subseteq S$, so $R = S/J$ and $I = \mathfrak{m}/J = \mathfrak{m}R$. Each $\text{in}(f)$ is homogeneous, so $\text{in}_I(J)$ is homogeneous and $\mathbb{V}(\text{in}_I(J))$ is a cone. We grade S by degree and define the k -algebra map

$$\phi : S \rightarrow \text{gr}_I R, \quad x_i \mapsto \bar{x}_i \in I/I^2.$$

Since S is free on the x_i this is well defined, and it is graded because the generators land in degree 1. It is surjective: $(\text{gr}_I R)_0 = R/I = k$, and as I is generated by the x_i , the piece I^n/I^{n+1} is spanned by degree- n monomials in the $\bar{x}_i$, i.e. by $\phi(S_n)$.

We claim $\ker \phi = \text{in}_I(J)$, degree by degree. Under $S_n \cong \mathfrak{m}^n / \mathfrak{m}^{n+1}$, the degree- n part of ϕ is the natural surjection

$$\mathfrak{m}^n / \mathfrak{m}^{n+1} \twoheadrightarrow I^n / I^{n+1} = (\mathfrak{m}^n + J) / (\mathfrak{m}^{n+1} + J) \cong \mathfrak{m}^n / (\mathfrak{m}^{n+1} + (J \cap \mathfrak{m}^n)),$$

whose kernel is $((J \cap \mathfrak{m}^n) + \mathfrak{m}^{n+1}) / \mathfrak{m}^{n+1}$. A degree- n form g lies in this kernel iff $g \equiv h \pmod{\mathfrak{m}^{n+1}}$ for some $h \in J \cap \mathfrak{m}^n$; writing $h = h_n + (\text{higher order})$, this says $g = h_n$, which is $\text{in}(h)$ when $\text{ord}(h) = n$ (and 0 otherwise). Conversely each such $\text{in}(h)$ is in the kernel. Finally, because $\text{gr}_{\mathfrak{m}} S \cong S$ is a domain, orders add ($\text{in}(g)\text{in}(h) = \text{in}(gh)$), so the degree- n part of the ideal $\text{in}_I(J)$ is exactly $\text{span}_k\{\text{in}(h) : h \in J, \text{ord } h = n\}$. Hence $\ker \Phi = \text{in}_I(J)$ and

$$\text{gr}_I R \cong S / \text{in}_I(J).$$

For $f \in J$ write $f = \text{in}(f) + (\text{higher order})$ with $\text{ord } f = d$. If $[v]$ is a limit of secant directions, say $p_n = t_n v_n \rightarrow 0$ with $t_n \rightarrow 0$ and $v_n \rightarrow v$, then

$$0 = f(t_n v_n) = t_n^d (\text{in}(f)(v_n) + t_n(\cdots));$$

dividing by t_n^d and letting $n \rightarrow \infty$ gives $\text{in}(f)(v) = 0$. As this holds for all $f \in J$, every limiting direction lies in $\mathbb{V}(\text{in}_I(J))$, so $TC_0(X) \subseteq \mathbb{V}(\text{in}_I(J))$.

For the reverse inclusion it is cleanest to use the blowup: by the isomorphism above, the projectivized cone $\mathbb{P} \mathbb{V}(\text{in}_I(J)) = \text{Proj}(S / \text{in}_I(J)) = \text{Proj}(\text{gr}_I R)$ is exactly the exceptional fibre $\pi^{-1}(0)$ of $\text{Bl}_0 X \rightarrow X$ (§3). Properness of π identifies the points of this fibre with the limiting secant directions, giving $\mathbb{V}(\text{in}_I(J)) \subseteq TC_0(X)$. Over the algebraically closed field k the two cones therefore agree as sets. $\square$

Intuition, and what the proof really needs

Two ideas drive the theorem, one algebraic and one geometric, and it helps to keep them apart.

The *algebraic engine* is the identity $\text{gr}_I R \cong S / \text{in}_I(J)$. The polynomial ring S is its own associated graded $\text{gr}_{\mathfrak{m}} S$, so Φ is just the map induced on associated graded rings by the quotient $S \twoheadrightarrow R$; passing to gr replaces each $f \in J$ by its lowest-degree part $\text{in}(f)$. The only real subtlety — and it is the whole content — is that the kernel is generated by the initial forms of *all* elements of J , not merely of a chosen generating set (recall Remark 2.7: $\text{in}_I(J)$ can be strictly larger than $(\text{in} f_1, \dots, \text{in} f_s)$). Computing the kernel degree by degree, as above, is the actual work; the rest is bookkeeping.

The *geometric picture* is that $\text{in}(f)$ is the lowest-order term of f at 0, so $\mathbb{V}(\text{in}(f))$ is the best homogeneous approximation to $\{f = 0\}$ at the origin — its tangent cone. Intersecting over all $f \in J$ gives the tangent cone of X .

Two honest caveats about the proof:

- **The two inclusions are not symmetric.** That a limiting secant direction kills every $\text{in}(f)$ is the easy Taylor expansion above. The converse — that *every* point of $\mathbb{V}(\text{in}_I(J))$ is realized as a limit of secant lines — is the substantive direction, and it is *not* “the same argument backwards.” It follows from the blowup being proper (its exceptional fibre is $\text{Proj} \text{gr}_I R$), or, over $\mathbb{C}$, from a curve-selection/limit argument. The sequential “ $p_n \rightarrow 0$ ” language also presupposes a topology: over $\mathbb{C}$ it is literal, but over a general algebraically closed field one replaces limits by the Zariski topology and the valuative criterion — or simply *defines* $TC_0(X)$ algebraically as $\text{Spec } \text{gr}_I R$.
- **Set versus scheme.** $\text{in}_I(J)$ need not be radical, so the equality $TC_0(X) = \mathbb{V}(\text{in}_I(J))$ is an equality of *sets* (it pins down $\sqrt{\text{in}_I(J)}$). The “moreover” — coordinate ring = $\text{gr}_I R = S / \text{in}_I(J)$

— is the finer scheme-theoretic statement, and it is the one carrying the multiplicity. For the node below, $\text{in}_I(J) = (y^2 - x^2)$ is radical, so the distinction is invisible there; in general (e.g. a cusp) it is not.

Example 4.1. For the nodal cubic $X = \mathbb{V}(y^2 - x^3 - x^2)$, with $f = y^2 - x^3 - x^2$, the lowest-degree part is $\text{in}(f) = y^2 - x^2 = (y - x)(y + x)$. Since $J = (f)$ is principal and orders add in S , $\text{in}_I(J) = (\text{in } f) = (y^2 - x^2)$. Hence

$$TC_0(X) = \mathbb{V}(y^2 - x^2) \subseteq \mathbb{A}^2,$$

the union of the lines $y = x$ and $y = -x$: the two tangent directions at the node. Its coordinate ring is $k[x, y]/(y^2 - x^2) \cong \text{gr}_I R$, and Proj of it is the two points $[1 : 1], [1 : -1] \in \mathbb{P}^1$, matching the exceptional set of §3. (The Hilbert function $1, 2, 2, 2, \dots$ of $\text{gr}_I R$ records multiplicity 2 — the node is a double point.)